

|           |          |                                                                                                                                                                                                                                                                                                 |
|-----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |          | Transition from $E_3$ to $E_2$ gives the same energy.                                                                                                                                                                                                                                           |
| <b>29</b> | <b>B</b> | The p.d. through which the electrons are accelerated will increase the kinetic energy of the electrons. This will increase the maximum energy of X-rays emitted, reducing the cut-off wavelength of the spectrum. The characteristic wavelengths remain unchanged as the target is not changed. |

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**30**

**A**

The correct answer describes a piece of evidence that uniquely identifies  $\alpha$ -radiation